Spam Email Detection using NLP and Machine Learning

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1 Problem Statement

Unsolicited and potentially harmful spam messages pose a significant challenge in digital communication[cite: 59]. The objective of this project is to develop an efficient machine learning-based system capable of automatically classifying incoming messages as *spam* or *ham* (legitimate)[cite: 60]. The project specifically aims to:

- Design a machine learning model for accurate detection[cite: 63].
- Apply NLP techniques for effective text preprocessing and feature extraction[cite: 64].
- Evaluate model performance using classification metrics[cite: 65].

2 System Architecture

The methodology follows a structured pipeline from raw data to classification[cite: 69]:

1. **Data Collection:** Loading the dataset (spam.csv) containing labeled SMS/email messages[cite: 71].
2. **Data Cleaning:** Removing unnecessary columns (Unnamed: 2, 3, 4), handling duplicates, and encoding targets (Ham=0, Spam=1)[cite: 73, 74].
3. **EDA:** Statistical analysis of character, word, and sentence counts to identify distribution patterns[cite: 76, 78].
4. **Preprocessing:** Converting text to lowercase, tokenization, removing stopwords/punctuation, and applying Porter Stemming[cite: 85, 86, 87, 88].
5. **Feature Extraction:** Utilizing TF-IDF Vectorization to transform text into numerical feature matrices[cite: 21].
6. **Modeling & Evaluation:** Training multiple classifiers and comparing results based on Accuracy and Precision[cite: 23, 24].

3 Tools and Libraries

The project was implemented in Python using the following specialized tools[cite: 1, 72, 78, 26]:

- **Pandas & NumPy:** For data manipulation and loading.
- **NLTK:** For tokenization and stemming.
- **Scikit-Learn:** For feature extraction (TF-IDF) and machine learning models.
- **Matplotlib & Seaborn:** For data visualization (pie charts and histograms).

4 Models Evaluated

The following supervised learning algorithms were implemented for comparative analysis:

- Multinomial Naïve Bayes (MNB)
- Bernoulli Naïve Bayes (BNB)
- Gaussian Naïve Bayes (GNB)
- Support Vector Machine (SVM)
- Random Forest Classifier (RFC)
- Logistic Regression (LR)
- Decision Tree K-Nearest Neighbors (KNN)

5 Results and Performance

Upon evaluation, Naïve Bayes variants exhibited superior performance for text classification. Specifically, the **Bernoulli Naïve Bayes** achieved the highest accuracy and reliability among all tested algorithms.

Table 1: Model Performance Comparison

Algorithm	**Accuracy**	**Precision**
Bernoulli NB	*Highest*	*1.00*
Multinomial NB	0.97	1.00
Random Forest	0.97	0.98
SVC	0.97	0.97

The Exploratory Data Analysis (EDA) revealed that spam messages generally contain a higher number of characters and words compared to ham messages[cite: 37].

6 Conclusion

This project demonstrates that combining NLP-based TF-IDF feature extraction with supervised models can effectively filter spam. The precision score was prioritized over accuracy to ensure no legitimate emails (Ham) are misclassified as Spam.